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RESEARCH ARTICLE



Grant counts, funding amount, and academic productivity: causal evidence from the NSFC

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ABSTRACT

This study explores the causal relationship between research funding and academic productivity, focusing on the National Natural Science Foundation of China (NSFC). Grounded in Resource Dependence Theory and Cumulative Advantage Theory, the research distinguishes between two key funding mechanisms: grant count (indirect) and funding amount (direct). Using longitudinal data from a leading Chinese polytechnic university covering the period 2005–2019, and applying an instrumental-variable approach, we find that competitive research funding has a significant positive effect on both the quantity and quality of scientific publications. Specifically, grant count supports sustained productivity by enabling project continuity, facilitating collaboration, and accumulating experience, particularly among early-career scholars. In contrast, funding amount provides the essential resources needed to execute more complex, high-impact research projects. Grant count emerges as the stronger driver of cumulative academic output, especially for young faculty, by maintaining stable research pipelines and fostering long-term growth. These findings underscore the critical role of funding systems that balance continuous smaller grants with larger-scale investments, ensuring academic productivity, research capacity, and career progression across different academic stages.

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Research funding; grant count; funding amount; academic productivity; career development

Introduction

Research funding has become a defining instrument in shaping the productivity, collaboration, and global competitiveness of universities. Across national systems, governments have institutionalized competitive grant schemes, such as the U.S. National Science Foundation (NSF), the Australian Research Council (ARC), and the UK Research and Innovation (UKRI), to promote scientific advancement and reinforce national innovation capacity (Abramo and D'Angelo 2023; Bloch, Graversen, and Pedersen 2014). Within this global landscape, the National Natural Science Foundation of China (NSFC) has emerged as China's primary competitive funding agency, distributing substantial resources to support basic research and technological innovation (Horta, Cattaneo, and Meoli 2018; Hu 2020; Zhou, Cai, and Lyu 2020). The NSFC budget for 2025 has increased significantly, with CNY 3.95 billion allocated to the Natural Science Fund, an 8.65% rise from 2024. This budget is primarily used to fund various research programs, such as general projects, youth science fund,

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regional science funds, and innovation groups (NSFC 2025). The NSFC's rapid budget expansion and its integration into university performance evaluations have made grant acquisition a critical determinant of both institutional reputation and individual career promotion within China's higher education system.

Career development for research academic faculty is the cumulative process through which scholars strengthen their research productivity, secure competitive funding, build collaborative networks, and advance within academic hierarchies shaped by performance-based evaluation and resource allocation systems (Abramo and D'Angelo 2023; Auranen and Nieminen 2010; Álvarez-Bornstein and Bordons 2021; Baldwin and Blackburn 1981). Funding provides not only the material and human resources, including laboratories, research assistants, and research infrastructure, necessary to sustain inquiry, but also serves as a symbolic signal of competence and prestige. Empirical studies demonstrate that externally funded academics tend to publish more frequently, achieve higher citation impact, and attain greater visibility in academic networks (Lepori, Jongbloed, and Hicks 2023; Pietilä 2025; Sinatra et al. 2016). Nevertheless, scholarly interpretations of how funding shapes productivity diverge. One perspective suggests that insufficient funding endangers academic careers and undermines research continuity (Fan and Qin 2025; Laudel 2006), while another contends that competition for scarce resources incentivizes efficiency, strategic focus, and innovation (Ayoubi, Pezzoni, and Visentin 2019; Wang, Thijs, and Glänzel 2015). Despite this long-standing debate, most empirical research remains correlational, leaving the causal relationship between research funding and scholarly output insufficiently understood. Moreover, few studies distinguish whether the funding amount or grant count exerts a stronger influence on academic productivity.

This study addresses these theoretical and empirical gaps by integrating Resource Dependence Theory (RDT) (Pfeffer and Salancik 1978) with the logic of Cumulative Advantage (CA) (Merton 1968). From an RDT perspective, research grants constitute critical external resources that directly reduce environmental uncertainty and enhance scholars' autonomy, while cumulative advantage emphasizes recursive feedback loops through which early funding success amplifies subsequent opportunities for recognition and resource acquisition indirectly (Baldwin and Blackburn 1981; Schein 1992). Together, these perspectives conceptualize funding as both material capital and symbolic capital that sustain productivity and shape stratified academic outcomes. Within this framework, the study examines how the grant count and the funding amount exert distinct causal effects on the quantity and quality of academic productivity, thereby providing a more granular understanding of funding-productivity dynamics in higher education.

Empirically, the analysis employs longitudinal bibliometric and administrative data (2005–2019) from a leading polytechnic university in China, integrating NSFC archives with individual faculty publication records. The dataset contains detailed information on project type, duration, and funding amount, enabling the estimation of causal effects through an instrumental-variables (IV) approach that exploits exogenous variation in discipline-year project approval rates. This strategy addresses endogeneity arising from self-selection and prior performance bias. By situating the case within China's competitive funding ecosystem, the study offers insights relevant to broader discussions of how national funding architectures influence academic productivity and institutional stratification across higher education systems worldwide.

The study makes three primary contributions. First, it advances theory by integrating RDT and CA perspectives to explain the direct and indirect effects of research funding, demonstrating how continuity and scale jointly structure research outcomes. Second, it introduces a robust empirical design and integrated dataset that provide causal evidence on how grant count and funding amount, as indirect and direct mechanisms, magnitude influence both the quantity and quality of academic output. Third, it contributes policy insights for designing balanced funding portfolios that combine recurrent smaller grants with larger, targeted investments, to promote sustained research momentum across career stages. The following sections present the theoretical framework, methodology, results, and implications for higher education policy and practice.

Literature review

The relationship between research funding and scientific productivity has long been central to higher education and science policy debates, yet its underlying mechanisms and causal dynamics remain contested. Two theoretical perspectives, the RDT and the CA Theory, provide complementary explanations for how funding shapes academic output. According to RDT, academic productivity is constrained by resource scarcity and uncertainty surrounding sustained research support. External funding mitigates these limitations by providing the financial, infrastructural, and temporal means required for long-term inquiry, allowing scholars to pursue projects that exceed institutional budgets (Pfeffer and Salancik 1978). Competitive grants thus enhance academic autonomy and enable research continuity by supplying essential materials, personnel, and protected time for scholarship. Empirical studies affirm this resource-based logic that access to competitive grants promotes sustained productivity growth, collaborative capacity, and innovation (Bloch, Graversen, and Pederesen 2014; Ou et al. 2024).

At the same time, research funding operates as a form of symbolic and reputational capital, reinforcing inequalities in recognition and influence. Merton's (1968) theory of cumulative advantage suggests that early success in obtaining grants generates self-reinforcing cycles of visibility and opportunity, whereby recognized scholars become more likely to attract subsequent funding. Later empirical work shows that this 'Matthew effect' persists in modern funding systems. Early recipients of competitive grants enjoy disproportionate access to future awards and collaborations (Bol, de Vaan, and van de Rijt 2018). Bourdieu's (1988) sociological framework further explains how academic capital, economic, social, and symbolic, accumulates through these recursive processes, legitimating hierarchies of prestige and authority. Together, RDT and CA explain why funding functions not only as a material enabler but also as a mechanism of symbolic differentiation. Few studies, however, have empirically integrated these mechanisms to evaluate their joint effects on research productivity. This study adopts both perspectives to conceptualize funding as simultaneously resource-enabling and status-reinforcing, shaping the dynamics of academic performance over time.

Understanding the causal relationship between research funding mechanisms and scientific productivity has become increasingly important in higher education policy research (Liu, Qin, and Li 2025). Competitive funding systems have evolved into core instruments of governance, designed to stimulate innovation, accountability, and performance across national research systems (Banal-Estañol et al. 2023; Lepori, Jongbloed, and Hicks 2023). These mechanisms rely on peer review and competition to allocate resources efficiently while incentivizing excellence and measurable outcomes. Empirical evidence consistently shows that funded academics outperform unfunded peers in terms of both publication quantity and impact (Ebadi and Schiffauerova 2016; Horta, Cattaneo, and Meoli 2018). However, such relationships are often inferred rather than causally demonstrated. Many studies rely on cross-sectional designs that cannot rule out selection bias, since already productive researchers are more likely to obtain grants (Fan and Qin 2025). Ding and Moreira (2025) demonstrate that after controlling for prior performance, the direct causal effect of winning a grant from NSF is significantly reduced, highlighting the need for longitudinal designs and quasi-experimental methods (Benavente et al. 2012).

In China, the NSFC represents the country's most important competitive funding mechanism in science and engineering. Its structure, combining peer review, project-type differentiation, and career-stage targeting, makes it a paradigmatic case for analyzing funding mechanisms and their productivity effects (Hu 2020; Zhou, Cai, and Lyu 2020). The NSFC has been instrumental in expanding China's research capacity, enhancing international collaboration, and elevating visibility in global science. Yet most studies of the NSFC remain descriptive, focusing on publication counts rather than on the causal mechanisms linking competitive funding to productivity growth (Ebadi and Schiffauerova 2016; Hu 2020; Ou et al. 2024; Tian, Zhang, and Chen 2024; Zhou, Cai, and Lyu 2020). Addressing this gap, the present study employs longitudinal faculty-level data to explore how competitive funding affects the trajectories of research output and influence.

While the positive association between funding and productivity is well established, the dimensions of funding that matter most remain unclear. Most studies treat funding as a binary condition, that is funded versus unfunded, overlooking the distinction between grant count and funding amount (Ayoubi, Pezzoni, and Visentin 2019; Benavente et al. 2012; Bloch, Graversen, and Pedersen 2014; Ebadi and Schiffauerova 2016; Hottenrott and Lawson 2017). These represent distinct causal mechanisms that may shape productivity differently. A higher grant count signals continuity and momentum that repeated awards sustain project pipelines, stabilize teams, and support cumulative collaboration over time (Álvarez-Bornstein and Bordons 2021; Bloch, Graversen, and Pedersen 2014). By contrast, a larger funding amount indicates depth and capacity, enabling capital-intensive projects that yield fewer but higher impact outputs (Hottenrott and Lawson 2017). Recent bibliometric analyses highlight this contrast. A study finds that while most STEM publications acknowledge funding, relatively few specify multiple awards or cumulative funding levels, leaving uncertain how continuity versus scale drives productivity (Tian, Zhang, and Chen 2024). Bone and Sherbon (2024) similarly report that frequent smaller grants foster sustained productivity, whereas large one-off awards concentrate resources but generate higher impact results.

This ongoing debate lies at the intersection of quality and quantity in research funding policy (Anninos 2014). If smaller, recurrent grants foster steady publication output, systems prioritizing broad participation may yield greater aggregate productivity. Conversely, if large-scale awards deliver proportionally higher scholarly influence, concentrating resources may be more effective for achieving high-quality breakthroughs. Existing findings remain mixed, often neglecting interactions between grant frequency and scale across career stages and institutional contexts. To address this theoretical and empirical gap, the present study explicitly differentiates between grant count and funding amount, examining their differential and joint effects on scientific productivity among Chinese academics in polytechnic universities.

Theoretical framework and hypotheses

In this study, we distinguish between two analytically distinct funding mechanisms. Funding amount captures a direct resource mechanism, whereby larger grants immediately expand researchers' material capacity, consistent with Resource Dependence Theory. Grant count captures an indirect, cumulative mechanism, whereby repeated grant acquisition sustains research continuity and builds long-term capacity over time, as emphasized in Cumulative Advantage theory. Importantly, 'indirect' denotes a temporally cumulative process rather than statistical mediation. Within China's NSFC system, characterized by annual competitive cycles and hierarchical evaluation structures, these mechanisms coexist and interact as distinct yet interdependent pathways through which research funding shapes faculty productivity. Grounded in this distinction, the study examines how funding amount and grant count differentially influence academic output.

H1: Receiving competitive research funding has a positive causal effect on the quantity and quality of scientific publications by university faculty.

H2: Grant count positively affects publication quantity and quality by supporting sustained project continuity and the accumulation of experience and collaboration over time.

H3: Total funding amount positively affects publication quantity and quality by directly providing the financial and human resources necessary for research execution.

H4: Grant count, as an indirect mechanism, exerts a stronger influence on cumulative publication output than funding amount, which represents the scale-based direct mechanism.

Figure 1 conceptually depicts this dual-mechanism model. Funding amount represents the direct resource mechanism of RDT, whereas grant count represents the indirect accumulative mechanism of CA. Both channels lead to increased scientific productivity, but through different temporal and structural pathways. The framework thus positions funding as both an immediate enabler of

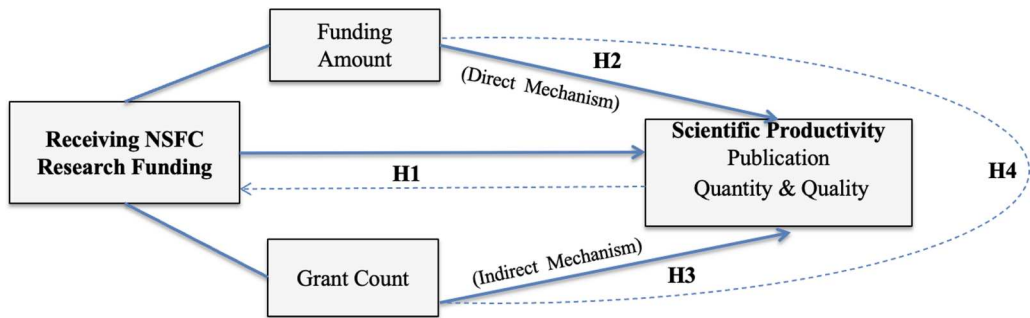


Figure 1. Dual mechanisms linking research funding and scientific productivity.

research and a long-term catalyst for cumulative academic development, central concerns in higher education policy and institutional performance systems.

Method

Research context and case description

This study investigates how research funding influences academic productivity through a longitudinal analysis of E University, a leading research-intensive institution in China. The university is a Double First-Class science and engineering university with over 2,000 faculty members and a consistent record of participation in the NSFC. Its disciplinary breadth and engagement in national funding schemes make it an analytically rich case for exploring how the frequency and scale of research funding affect scholarly productivity within China's evolving, performance-based higher education system. The period from 2005 to 2019 represents a critical phase of China's higher education transformation, marked by a shift from block funding to competitive, project-based allocation. During this time, E University experienced a steady increase in both the number of NSFC projects awarded and the total volume of grants received, reflecting the national emphasis on performance-oriented academic evaluation. Within this institutional environment, external grants not only provide material resources but also constitute a key symbol of academic prestige and legitimacy, shaping both career advancement and institutional reputation.

Descriptive statistics in Table 1 reveal a marked divide between funded and non-funded faculty at E University. During 2009–2019, 877 academics received at least one grant from the NSFC, averaging 1.285 grants and a total funding amount of CNY 730,000 (approximately USD 100,000). These faculty produced, on average, 11.5 publications, compared with 7.567 publications among their non-funded peers. Funded scholars also demonstrated stronger citation impact, measured by Category Normalized Citation Impact (CNCI) (2.937 vs. 2.341), higher journal visibility, measured by Journal Impact Factor (JIF) (4.114 vs. 3.502), and stronger journal-normalized performance, measured by Journal Normalized Citation Impact (JNCI). Differences in Average Percentile of papers (AP) further indicate variation in the relative standing of publications within their disciplinary and temporal reference sets. Although funded academics were generally more senior and better credentialed, the descriptive results also suggest a potential trade-off: faculty holding multiple grants exhibited higher publication volumes but slightly lower mean CNCI and JNCI. Overall, these patterns indicate a strong descriptive association between NSFC funding and academic productivity and visibility, while leaving open questions regarding the causal pathways and quality implications examined in the subsequent analysis.

Faculty profiles were obtained from E University's official website using structured web scraping, followed by manual verification of key fields such as academic background, rank, and disciplinary affiliation. These records were cross-validated with internal data obtained from the university's

Table 1. Descriptive analysis of indicators related to obtained NSFC and publications.

Variables	Obtained of NSFC Programs					Non-Obtained NSFC Programs				
	Observation	Mean	SD	Minimum	Maximum	Observation	Mean	SD	Minimum	Maximum
Grant count	877	1.285	0.596	1	6					
Funding amount (CNY 10,000)	877	73.786	87.565	8	1346					
Number of published Papers	632	11.500	13.081	1	107	1110	7.567	9.648	1	80
Number of collaborative Papers	632	4.5188	8.261	0	83	1110	1.075	3.614	0	50
Journal Normalized Citation Impact (JNCI)	632	0.990	0.712	0	10.780	1110	0.891	0.834	0	12.039
Journal Impact Factor (JIF)	632	4.114	2.170	0.3	16.257	1110	3.502	1.819	0.252	14.050
Category Normalized Citation Impact (CNCI)	632	2.937	5.295	0	50.145	1110	2.341	4.689	0	42.452
Article Percentile (AP).	632	44.242	19.580	1.0298	100	1110	52.236	22.169	0.7244	100
Gender	877	0.677	0.468	0	1	1110	0.604	0.489	0	1
Age Group	877	1.803	0.758	1	4	1110	2.089	0.807	1	4
Graduated Institution Type	877	1.609	0.652	1	3	1110	1.657	0.632	1	3
Education Level	877	3.985	0.121	3	4	1110	3.611	0.662	1	4
Discipline Type	877	1.609	0.959	1	4	1110	1.826	1.154	1	4
Employment Type	877	1.419	0.494	1	2	1110	1.258	0.438	1	2

Human Resources, Science and Technology, and Development and Planning offices, which provided verified information on employment status, promotion history, and participation in talent programs. Research funding data were retrieved from the NSFC's publicly accessible archives, which document project titles, applicants, award categories, approval dates, and total grant amounts. The datasets were integrated through a MySQL relational database using unique faculty identifiers to ensure consistency across data sources. Bibliometric data were sourced from the Web of Science (WoS), focusing on publications affiliated with E University. Name disambiguation and institutional verification procedures were conducted to ensure authorship accuracy.

The final panel dataset includes 1,156 full-time faculty members over a 15-year period, encompassing 17,134 verified SCI-indexed papers. This integration allows for a fine-grained examination of how individual funding trajectories relate to research output over time, providing an internally valid case for testing the mechanisms linking research capital and productivity.

Variable definition and measurement

Academic productivity, the dependent variable, is measured as the number of SCI-indexed publications per faculty member within each five-year career-stage window. This measure reflects the dominant indicator used in national research evaluations and allows comparability across disciplines and over time. Complementary indicators, including JIF, CNCI, AP, and JNCI, were also calculated within the same five-year career-stage window to capture output quality and visibility, though publication count serves as the primary outcome variable. CNCI measures the normalized average number of citations per paper, adjusted by discipline, publication year, and document type. AP measures the percentile of total citations for papers within the same discipline and publication year. JNCI provides a standardized metric for calculating citations of documents published in specific journals. JIF is calculated by the ratio of total citations obtained by papers published in a journal over the past two years to the number of academic works published in that journal in the same timeframe.

The two key independent variables capture distinct dimensions of research funding: grant count, defined as the total number of NSFC projects awarded to each faculty member within a career stage, and funding amount, representing the total monetary value of NSFC grants. These two measures operationalize the dual-mechanism theoretical model, where funding scale reflects a direct resource effect, while funding count captures an indirect accumulative effect associated with sustained learning and collaboration.

Several control variables were included to account for potential confounding influences at both the individual and institutional levels. Individual characteristics comprised gender (male = 1, female = 0), age group (25–35, 36–45, 46–60, and above 60 years). Pre-career and post-career controls included graduated institution type, education level, discipline type, and employment type, reflecting both formative academic experiences and current professional contexts that shape research performance. Education level is highest degree (associate = 1, bachelor = 2, master = 3, doctoral = 4), and graduated institution type (E university = 1, domestic universities = 2, and overseas universities = 3). Structural and institutional features included discipline type, measured by the faculty member's current departmental affiliation (science, engineering, medicine, and other disciplines), and employment type (research-oriented or teaching-oriented). These variables capture heterogeneity in career stage, educational background, and academic qualifications that may influence research productivity and access to funding. Year and discipline fixed effects were added to account for field-specific publication norms and temporal shifts in research policy. Data reliability was ensured through ORCID-based author identification, verification of institutional affiliations, and winsorization of the top 1% of funding observations to reduce the influence of extreme values. Together, these controls enhance the robustness of the estimation and ensure that observed relationships reflect the net effect of research funding on academic performance rather than underlying demographic or institutional heterogeneity.

Analytical design and IV identification

A key methodological challenge in estimating the effect of research funding on academic productivity concerns endogeneity. Faculty who are already productive or institutionally supported are systematically more likely to secure competitive grants, while receiving such funding enhances future productivity by expanding research capacity, collaborations, and visibility. This two-way causality, compounded by unobserved factors such as disciplinary reputation and departmental resources, risks biasing ordinary least squares (OLS) estimates and conflating selection effects with genuine causal influence.

To address this issue, the study adopts an instrumental variable (IV) strategy that exploits exogenous variation in the NSFC funding environment. It directly tests whether funding continuity (grant count) or funding intensity (funding amount) exerts greater influence on the quantity and quality of scholarly output. Because the NSFC allocates grants through discipline-based review panels whose approval rates fluctuate annually according to national priorities, budget allocations, and field-specific demand, these variations generate quasi-random differences in funding opportunities across years and disciplines. Such differences serve as a natural source of exogenous variation independent of individual faculty characteristics. This integration allows for a fine-grained examination of how individual funding trajectories relate to research output over time, providing an internally valid case for testing the mechanisms linking research capital and productivity.

Two potential instruments were considered that the discipline-year project approval rate (IV1) and the funding approval rate (IV2), both defined within each discipline-year cell of the NSFC database. The project approval rate captures the probability of a successful application, while the funding approval rate reflects the proportion of approved to requested funds. For each faculty member, these indicators were matched to their project's discipline, type, and year of application. Where multiple project types were obtained within a career stage, approval rates were averaged to construct a composite measure of funding accessibility.

To identify valid instruments for subsequent causal estimation, a series of funding-level ordinary least squares regressions were conducted (see Appendix Table A1). Model 1 examines the association between the total funding amount and the number of grants, revealing a strong positive relationship. Each additional grant corresponds to an average increase of CNY 81,384 in total funding ($p < .01$). Models 2 and 3 test the predictive strength of the discipline-year project approval rate (IV1) for funding outcomes. The results show that project approval rate is a statistically strong predictor of both funding amount and grant count, satisfying the relevance criterion for a valid instrument. In contrast, Models 4 and 5 indicate that the funding approval rate (IV2) lacks explanatory power, showing no significant association with either outcome. These findings demonstrate that the discipline-year IV1 captures systematic, quasi-exogenous variation in funding opportunities across disciplines and years, whereas the funding approval rate fails to meet the relevance condition. Accordingly, the discipline-year project approval rate was retained as the preferred instrumental variable in the subsequent analysis.

Results

Causal effects of research funding: frequency and scale mechanisms

This section examines the effects of research funding through two analytically distinct mechanisms: grant frequency, capturing a cumulative and indirect pathway, and funding scale, capturing a direct resource-based pathway. We first summarize baseline associations and then focus on instrumental-variable estimates to identify causal effects. To improve clarity and focus, we prioritize instrumental-variable estimates in the main text, while detailed first-stage regressions, extended model specifications, and robustness analyses are reported in the Online Appendix.

Table 2. IV regression of the grant count on the quantity and quality of academic papers.

Variable	Number of published Papers	CNCI	AP	JNCI	JIF
Grant count	22.790*** (4.18)	4.204** (2.12)	-41.995*** (3.71)	4.618*** (3.75)	0.836*** (2.74)
Demographic characteristics controls	Yes	Yes	Yes	Yes	Yes
Pre-career and post-career controls	Yes	Yes	Yes	Yes	Yes
Disciplines controls	Yes	Yes	Yes	Yes	Yes
Observation	626	626	626	626	626
Adjusted R ²	0.096	0.293	0.153	0.079	0.079

Note: Robust t-statistics in parentheses; ***p<0.01, **p<0.05, *p<0.1

Baseline ordinary least squares (OLS) estimates indicate a positive association between grant count and funding amount in (see Appendix [Tables A2](#) and [A3](#)). On average, each additional NSFC grant is associated with 6.093 additional publications over a five-year period, net of demographic, career-stage, and disciplinary controls. Modest improvements are observed for AP and JIF, while effects on JNCI and CNCI are small or statistically insignificant. These OLS patterns suggest that repeated grant participation is linked to sustained research activity, but they may be influenced by selection effects and therefore serve primarily as descriptive benchmarks.

IV estimates provide stronger evidence for a causal continuity mechanism associated with grant frequency. Using discipline-year project approval rates as an instrument, first-stage results indicate strong relevance and second-stage estimates in [Table 2](#) show that each additional grant increases publication output by approximately 22.79 articles over five years, nearly four times larger than the corresponding OLS estimate. Repeated funding is also associated with substantial gains in research visibility and placement, including increases in JNCI (+4.618) and JIF (+.836), as well as a marked improvement in AP (-41.995), indicating movement toward higher ranked journals. Taken together, these results suggest that grant frequency exerts a strong indirect effect by sustaining research continuity and cumulative capacity, pushing researchers closer to the upper tiers of global publication distributions.

Funding amount operates through a different channel. Baseline OLS estimates are reported in the Online Appendix and serve as descriptive benchmarks. The IV estimates in the main text reveal stronger causal effects in [Table 3](#). Specifically, each additional CNY 10,000 corresponds to a marginal increase of 0.101 publications and is associated with statistically significant improvements in JNCI, JIF, and AP (-0.186). These findings are consistent with a direct resource mechanism, whereby greater funding scale expands overall research capacity and supports more ambitious research activity. Although marginal effects at the unit level are small, their cumulative implications for larger projects are non-trivial.

The contrast between OLS benchmarks (Appendix [Tables A2](#) and [A3](#)) and IV estimates ([Tables 2](#) and [3](#)) underscores that conventional models substantially underestimate the returns to competitive research funding. More importantly, the results clarify the distinct roles of funding mechanisms within the NSFC system. Grant frequency primarily operates through an indirect, cumulative pathway that sustains productivity and research momentum, while funding scale operates

Table 3. IV regression of funding amount on the quantity and quality of academic papers.

Variable	Number of published Papers	CNCI	AP	JNCI	JIF
Amount of funding	0.101*** (5.56)	0.019** (2.25)	-0.186*** (4.75)	0.020*** (5.13)	0.004*** (3.04)
Demographic characteristics controls	Yes	Yes	Yes	Yes	Yes
Pre-career and post-career controls	Yes	Yes	Yes	Yes	Yes
Disciplines controls	Yes	Yes	Yes	Yes	Yes
Observation	626	626	626	626	626
Adjusted R ²	0.245	0.198	0.151	0.143	0.161

Note: Robust t-statistics in parentheses; ***p<0.01, **p<0.05, *p<0.1

through a direct capacity-expanding pathway that enhances output visibility and placement. The coexistence of these mechanisms reflects the tiered structure of the NSFC funding system and helps explain how competitive funding simultaneously supports broad participation and high-impact research trajectories across career stages.

Robustness, sensitivity, and interpretive insights

Robustness diagnostics confirm that the main causal estimates are stable and statistically credible. Across all instrumental-variable models, first-stage F-statistics exceed the conventional threshold of 10, and Kleibergen-Paap Wald statistics surpass Stock-Yogo critical values, ruling out weak-instrument concerns. Hansen J-tests fail to reject the null of instrument exogeneity, supporting the validity of discipline-year approval rates as instruments. Alternative specifications, including the exclusion of outliers, clustering standard errors by discipline, and median regression, yield substantively similar coefficients. Re-estimations using alternative quality indicators, such as CNCI and JIF, further confirm that the positive causal effects of both grant count and funding amount are robust across outcome measures. Placebo tests using lagged approval rates outside active funding cycles produce null effects, suggesting that the estimated impacts are not driven by underlying disciplinary trends.

As an additional robustness check, we examine collaborative research output, measured by the number of co-authored publications. The instrumental-variable results, reported in the Appendix [Table A4](#), show that grant count has a strong and statistically significant positive effect on collaborative publication output, while funding amount also exhibits a positive and statistically significant effect with a smaller marginal magnitude (approximately 0.076 collaborative publications per additional CNY 10,000). These findings are presented as robustness evidence rather than as primary outcomes and align closely with the mechanism-based interpretation advanced in the main analysis.

Interpreting the results through the lens of local average treatment effects provides further insight. The sizable IV coefficients reflect impacts among ‘complier’ scholars, those whose funding outcomes vary with discipline-level approval rates. These individuals, typically situated at mid-career stages, combine established research capacity with continued resource dependence. Their heightened responsiveness indicates that incremental increases in funding accessibility generate particularly large returns in both individual productivity and collaborative output. This heterogeneity refines cumulative advantage arguments by highlighting that sustained opportunity, rather than initial privilege alone, is central to long-term research development.

Overall, the results depict a coherent pattern in which competitive research funding in China operates through dual yet complementary logics. The direct resource mechanism translates funding scale into expanded research capacity, while the cumulative continuity mechanism converts repeated funding into sustained productivity and collaborative momentum. Their interaction, reinforced by the stratified structure of the NSFC system, helps explain how funding simultaneously consolidates institutional prestige and broadens participation. In doing so, the analysis offers both robust causal evidence and theoretical clarity regarding how funding architectures shape academic trajectories in contemporary higher education.

Discussion

This study examined how competitive research funding shapes academic productivity and professional advancement in a performance-driven higher education system. After addressing endogeneity, we find large causal effects of NSFC support on both publication quantity and field-normalized impact, and we clarify that funding works through two distinct mechanisms, grant count and funding amount, whose salience varies by career stage. Framed by the RDT (Pfeffer and Salancik 1978) and the logic of CA (Merton 1968), the results align China’s case with international

evidence on how funding architectures can simultaneously drive performance and stratify opportunity (Gläser and Laudel 2016; Whitley, Gläser, and Engwall 2010).

Although most previous studies have consistently found a positive association between research funding and academic output (Banal-Estañol et al. 2023; Ohniwa, Takeyasu, and Hibino 2023), our results indicate that this relationship has been systematically underestimated. After addressing the endogeneity between funding and performance, the causal impact of funding on both the quantity and quality of scholarly publications increases substantially. Funded academics not only publish more but also produce work with higher citation impact, measured through CNCI and JIF. These results confirm that research funding is not merely a correlate of productivity but a determinant of it, one that expands overall research capacity and facilitates sustained engagement in research activity.

In China's performance-based evaluation environment, where NSFC outcomes directly influence promotion and university ranking (Hu 2020), research grants also function as a form of symbolic legitimacy. This reinforces Gläser and Laudel's (2016) argument that state-steered competitive systems institutionalize academic hierarchies by equating funding success with academic value. Our findings thus align with the growing international evidence that research funding transforms not only what academics can do but also how their contributions are recognized and rewarded. The underestimation of funding's role in prior studies likely reflects methodological limitations, by controlling for unobserved productivity traits, this study isolates funding's true causal power in driving both academic productivity and career development.

Across the NSFC portfolio, the hierarchy is consistent that frequency underpins growth, learning, and stability; scale magnifies already-built capacity. In our estimates, one additional grant is associated with a sizable increase in publication output, while an average-sized grant yields a comparable gain when continuity is considered. Read together, these figures indicate that a run of smaller projects can match or even surpass the output of a single large project once continuity effects are taken into account. This quantitative contrast frames the mechanism-level interpretation below. International comparative work reaches similar conclusions about the superiority of stable support over isolated injections for building durable research capacity (Auranen and Nieminen 2010), and classic accounts of cumulative advantage explain why a sequence of small wins tends to be more visible and more convertible into further success than a single windfall (Merton 1968).

For early-career scholars, the Young Scientists Fund (about CNY 200,000 per project for researchers under 35) makes the logic particularly clear that funding frequency outweighs scale. A succession of youth grants typically yields larger gains in publication counts and journal placement than a one-off larger award, because repeated funding enables learning-by-doing, deepens collaboration ties, and may also facilitate collaborative engagement as teams stabilize over time. In resource-dependence terms, recurring external inflows reduce uncertainty when vulnerability is highest, allowing fledgling teams to bridge review cycles and retain key personnel (Pfeffer and Salancik 1978). Studies of other national systems report the same pattern that smaller, widely distributed awards are effective at seeding participation and accelerating first publications among newcomers (Ohniwa, Takeyasu, and Hibino 2023).

For mid-career scholars (around ages 35–45), programs such as Excellent Young Scientists and Young Distinguished (about CNY 1–3 million) raise the upper bounds of research capacity but do so most productively when layered on prior frequency. Large awards finance instrumentation, personnel depth, and international collaboration; however, their marginal returns are muted if they arrive without an established run of earlier projects. Where prior continuity has already built stable teams, routines, and reputational footing, concentrated funding accelerates outputs and improves placement, an organizational-learning dynamic in which repeated projects create the absorptive capacity needed to deploy big infusions effectively (Banal-Estañol et al. 2023). In cumulative-advantage terms, this is the expected 'frequency-then-scale' ladder that early continuity expands eligibility for larger awards, which in turn amplify subsequent performance (Merton 1968). In systems where evaluation tightly couples grants to careers, the stage-contingent payoff

structure of this ladder is typically intensified (Gläser and Laudel 2016; Whitley, Gläser, and Laudel 2018).

Across these comparisons, grant count is the foundation: it sustains people, routines, and momentum, while funding amount is the amplifier once that foundation exists. Securing grants repeatedly over time is at least as important as securing larger budgets, especially before capacity is fully built. This 'frequency-first, scale-second' logic clarifies why early-career pipelines respond so strongly to recurring support and why mid-career peaks occur when sizeable awards arrive on top of established continuity. It also explains heterogeneous returns across indicators that continuity drives volume and stability, whereas scale, once capacity is in place, helps lift placement and field-normalized impact. The results align with international evidence that career-stage-appropriate, recurring support generates more durable performance than regimes relying primarily on sporadic, high-value infusions (Auranen and Nieminen 2010; Gläser and Laudel 2016; Ohniwa, Takeyasu, and Hibino 2023).

Integrating Resource Dependence and Cumulative Advantage theories helps explain the structural implications of these findings. From an RDT perspective, frequent small grants mitigate uncertainty and secure a stable flow of external resources, which in turn strengthen researchers' autonomy within institutions. Large grants, while resource-rich, also heighten dependence on external evaluators and funding agencies, embedding academics within hierarchies of accountability. These structural asymmetries produce a dual regime of continuity and concentration, one nurturing inclusion, the other consolidating prestige. The prioritization of salary and job security by young researchers, as highlighted by Horta and Dai (2025), underscores the need to consider the broader economic context and the incentives that drive researchers' career decisions.

Through the lens of CA, these dynamics become recursive, that scholars who secure early small grants gain publication momentum and visibility, which increase their likelihood of obtaining large-scale funding later. Once acquired, major awards elevate their reputational capital, making future success more probable. Such recursive cycles have been documented in multiple national contexts (Ebadi and Schifffauerova 2016), but our findings demonstrate that the effect is amplified in China, where NSFC success is directly linked to academic promotion and institutional evaluation. In this system, continuity-based funding supports participation and renewal, while concentration-based funding accelerates elite consolidation, reinforcing a two-tier academic structure.

While grounded in China's context, these patterns echo broader international trends in competitive research funding. Funding mechanisms increasingly intertwine resource allocation with symbolic reputation (Whitley, Gläser, and Laudel 2018), creating similar tensions between excellence and equity. However, the coupling in China is unusually tight. The NSFC outcomes determine not only research resources but also institutional rankings, career mobility, and disciplinary legitimacy. In China, by contrast, the NSFC remains the determinant of academic standing. For China and comparable systems, the policy implication is clear. The long-term sustainability of research performance depends on recognizing the temporal logic of academic growth. Frequency fosters continuity, mentoring, and skill formation; scale amplifies established excellence. Funding frameworks that circulate opportunity rather than concentrate privilege will not only enhance equity but also strengthen the overall resilience and creativity of academic systems. These results underscore the importance of designing hybrid funding architectures that balance continuity and concentration. In the UK or EU, diverse funding streams, philanthropic, industrial, or regional, diffuse this dependence (Banal-Estañol et al. 2023; Zacharewicz et al. 2023). Systems overly reliant on large competitive grants risk marginalizing early-career scholars and narrowing disciplinary diversity, while excessive fragmentation through micro-grants can dilute focus and efficiency.

Conclusion

This study demonstrates that competitive research funding functions not only as a financial mechanism but as a structuring force that shapes academic productivity, legitimacy, and professional

advancement. Drawing on RDT and the CA, it provides causal evidence that public funding enhances both the quantity and quality of scholarly output. Addressing endogeneity reveals that prior research has systematically underestimated funding's influence on academic performance, while distinguishing between grant frequency and funding scale clarifies the dual mechanisms through which funding sustains and amplifies research activity.

Empirically, evidence from the NSFC illustrates that frequency, smaller-scale projects promote growth, learning, and stability, enabling scholars to accumulate experience, sustain collaborations, and maintain continuity across review cycles. Large-scale, concentrated grants amplify established capacity by providing the financial and human resources needed for complex research and international collaboration. Together, these mechanisms produce a recursive pattern of career development, in which recurrent support builds the absorptive capacity required to leverage large-scale investments effectively, while concentrated funding accelerates achievement once reputational and organizational stability is established. For early- and mid-career faculty, the continuity afforded by frequent grants yields more durable productivity gains than one-off large awards, underscoring that predictable support is foundational to long-term academic growth.

Theoretically, these findings refine RDT by showing how steady inflows of modest funding strengthen institutional autonomy and buffer uncertainty, whereas concentrated awards heighten dependence on external evaluative structures. They also extend the logic of CA by revealing how continuous support compounds recognition and capability over time, reinforcing career momentum and institutional hierarchies. Competitive funding systems thus operate as both enablers of excellence and architects of inequality. They expand capability even as they embed structural stratification.

From a policy perspective, sustaining a balanced research ecosystem requires hybrid funding architectures that integrate continuity with concentration. Smaller, recurring grants are essential for nurturing early- and mid-career participation and maintaining disciplinary diversity, while selective large-scale programs remain critical for high-impact innovation. Systems that rely exclusively on concentrated elite funding risk amplifying inequities, whereas excessive fragmentation through micro-grants can dilute focus and efficiency. Effective design should therefore circulate opportunity rather than consolidate privilege.

Limitations and future directions

This study contributes to understanding how the structure and rhythm of research funding influence academic productivity within a performance-based higher education system. Nevertheless, several limitations should be acknowledged. First, the analysis relies on longitudinal data from a single leading Chinese research university. While this institutional focus enhances internal consistency and allows for rigorous causal identification, it limits the external validity of findings to other universities with different governance models or funding dependencies. Second, the study employs bibliometric indicators as the principal measure of research performance. Although these indicators provide standardized and replicable evidence of scholarly output, they may not fully capture complementary dimensions of academic work such as mentorship, promotion and academic awards. Third, despite the use of an instrumental-variable approach to address endogeneity, unobserved heterogeneity, particularly in departmental leadership or internal resource allocation, may still influence the estimated relationships between funding and research performance.

Future research should therefore extend the empirical scope through multi-institutional or cross-national datasets, enabling comparative analyses across diverse institutional and policy contexts. Integrating qualitative or mixed-method approaches would further reveal how academics experience and adapt to performance-based funding regimes. In addition, expanding performance metrics to include interdisciplinarity, societal impact, and educational innovation would provide a more holistic understanding of research effectiveness. Ultimately, the empirical evidence suggests

that the effectiveness of research funding depends not only on the magnitude of financial resources but also on the structure and rhythm of their distribution. As higher education systems worldwide evolve toward performance-based accountability, identifying the balance between continuity and concentration remains crucial for designing funding systems that sustain both excellence and equity.

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Appendix

Table A1. Instrumental variables regression results.

Variables	Funding amount	Grant count	Funding amount	Grant count	Funding amount
Grant count	81.384*** (11.56)				
Project approval rate (IV1)		-2.305*** (-4.70)	-544.950*** (-4.53)		
Funding approval rate (IV2)				-0.274 (-0.71)	-32.524 (-0.52)
Years control	YES	YES	YES	YES	YES
Observation	944	944	944	944	944
Adjusted R ²	0.396	0.051	0.141	0.029	0.081

Note: Robust t-statistics in parentheses; *** $p < .01$, ** $p < .05$, * $p < .10$.

Table A2. OLS regression of grant count on the quantity and quality of publications.

Variable	Number of published papers			CNCI	AP	JNCI	JIF
	Model 6	Model 7	Model 8	Model 9	Model 10	Model 11	Model 12
Grant count	7.059*** (5.88)	6.093*** (5.25)	6.093*** (5.58)	0.354 (1.50)	-4.061*** (-3.78)	0.090** (2.11)	0.473*** (3.54)
Demographic characteristics controls	No	Yes	Yes	Yes	Yes	Yes	Yes
Pre-career and post-career controls	No	No	Yes	Yes	Yes	Yes	Yes
Disciplines controls	No	No	Yes	Yes	Yes	Yes	Yes
Observation	676	676	676	676	676	676	676
Adjusted R ²	0.135	0.274	0.304	0.020	0.033	0.013	0.102

Note: Robust t-statistics in parentheses; *** $p < .01$, ** $p < .05$, * $p < .10$.

Table A3. OLS regression of funding amount on the number and quality of publications.

Variable	Number of published papers			CNCI	AP	JNCI	JIF
Funding amount	0.067*** (5.15)	0.056*** (4.62)	0.055*** (4.57)	-0.000 (-0.31)	-0.027*** (-3.40)	0.000 (1.64)	0.006*** (4.17)
Demographic characteristics controls	No	Yes	Yes	Yes	Yes	Yes	Yes
Pre-career and post-career controls	No	No	Yes	Yes	Yes	Yes	Yes
Disciplines controls	No	No	Yes	Yes	Yes	Yes	Yes
Observation	626	626	626	626	626	626	626
Adjusted R ²	0.246	0.333	0.358	0.018	0.031	0.010	0.139

Note: Robust t-statistics in parentheses; *** $p < .01$, ** $p < .05$, * $p < .10$.

Table A4. IV regression of grant count and funding amount on the number of collaborative papers.

Variable	First Stage Grant count	second stage Number of collaborative papers	First Stage Funding amount	second stage Number of collaborative papers
Grant count		16.889*** (4.33)		
Funding amount				0.076*** (5.86)
Project approval rate	-2.528*** (-4.53)		-562.090*** (-7.26)	
First stage Partial R2	0.150	.	0.180	
First stage F-stat	13.500	.	15.45	
Demographic characteristics controls	Yes	Yes	Yes	Yes
Pre-career and post-career controls	Yes	Yes	Yes	Yes
Disciplines controls	Yes	Yes	Yes	Yes
Observation	855	855	855	855
Adjusted R2	0.091		0.193	

Note: Robust t-statistics in parentheses; *** $p < .01$, ** $p < .05$, * $p < .10$.